

SAT Diagnostic Test (Elite AI)

Module 1: Reading and Writing

Question 1 (*Transitions - Easy*)

The heavy rainfall lasted for several consecutive days, flooding many low-lying neighborhoods. , hundreds of residents were forced to evacuate their homes. Which choice completes the text with the most logical transition?

- A) Nevertheless
- B) Consequently
- C) Meanwhile
- D) Instead

Question 2 (*Command of Evidence: Quantitative - Medium*)

A table shows vaccine efficacy by age: |Age|Efficacy| |---|---| |18-39|96%| |40-59|91%| |60-74|85%| |75+|72%| A public health official says the vaccine 'works equally well for all age groups.' Is this accurate?

- A) No, because the vaccine doesn't work.
- B) No. While efficacy is high across all groups, it declines meaningfully with age from 96% for the youngest to 72% for the oldest a 24-percentage-point difference that contradicts 'equally well.'
- C) Yes, because the vaccine works for everyone.
- D) Yes, because all groups are above 50%.

Question 3 (*Central Ideas and Details - Easy*)

A passage examines how the printing press not only made books cheaper but also standardized spelling, enabled the Protestant Reformation by making the Bible widely accessible, and accelerated the Scientific Revolution. The central idea is:

- A) Spelling is important.
- B) The printing press was invented by Gutenberg.
- C) Books became cheaper after the printing press.
- D) The printing press's impact extended far beyond publishing: by democratizing information, it catalyzed religious, scientific, and linguistic transformations that fundamentally reshaped Western civilization.

Question 4 (*Inferences - Medium*)

Researchers found that students who took handwritten notes remembered conceptual information better than those who typed notes, even though typists recorded more words. The researchers hypothesize that handwriting forces students to process and rephrase information rather than transcribing verbatim. What can be inferred?

- A) Students should never use laptops in class.
- B) Typing notes is always inferior to handwriting.
- C) Handwritten notes contain more information than typed notes.
- D) The cognitive effort required by handwriting may enhance understanding more than the speed advantage of typing.

Question 5 (Central Ideas and Details - Easy)

Sign language is a fully developed language with its own grammar, syntax, and vocabulary; it is not simply a translation of spoken language into hand gestures. American Sign Language (ASL), for example, has grammatical structures that differ significantly from English. Linguists recognize sign languages as natural languages that develop within deaf communities and evolve over time just like spoken languages. Which choice best states the main idea of the text?

- A) Sign languages are complete, independent languages with their own grammar, not mere translations of spoken languages.
- B) ASL is the most widely used sign language in the world.
- C) Deaf communities are unable to communicate without learning sign language.
- D) Sign language and spoken English share identical grammatical structures.

Question 6 (Transitions - Medium)

The study found that moderate coffee consumption was associated with a 20% lower risk of Type 2 diabetes. excessive consumption more than six cups daily was linked to increased anxiety and insomnia. Which transition is most logical?

- A) Therefore,
- B) For instance,
- C) In addition,
- D) However,

Question 7 (Command of Evidence: Textual - Easy)

Ecologist Dr. Wen argues that urban green roofs significantly reduce city temperatures. Which finding most directly supports this claim?

- A) Green roofs are aesthetically pleasing to most city residents.
- B) Plants in general require water and sunlight to grow.
- C) Some cities have passed ordinances requiring new buildings to include green spaces.

D) A study of 50 buildings with green roofs showed an average 3.5C reduction in surrounding air temperature compared to buildings with conventional roofs during summer months.

Question 8 (Inferences - Easy)

Dr. Smith's latest experiment involved growing tomato plants under two different lighting conditions. Group A received natural sunlight, while Group B was exposed exclusively to artificial LED light. After four weeks, the plants in Group A had grown an average of 15 centimeters, whereas those in Group B had grown an average of 8 centimeters. Which choice is most strongly supported by the text?

- A) The tomato plants grew more when exposed to natural sunlight than when exposed to artificial LED light.
- B) Natural sunlight provides essential nutrients that artificial LED light lacks.
- C) Dr. Smith expected the plants in Group B to grow taller than those in Group A.
- D) Artificial LED light is harmful and causes tomato plants to wither over time.

Question 9 (Command of Evidence: Quantitative - Hard)

A marine biologist studied coral bleaching events across four ocean regions: | Region | Water Temp Increase (C) | % Coral Bleached | Recovery Rate (%) | |---|---|---|---| | Caribbean | +1.2 | 48% | 22% | | Southeast Asia | +0.8 | 31% | 45% | | Great Barrier Reef | +1.5 | 62% | 15% | | Red Sea | +1.0 | 18% | 68% | Which conclusion is best supported by the data?

- A) Temperature increases have no effect on coral bleaching, since the Red Sea shows low bleaching despite a 1.0C increase.
- B) Recovery rates are determined solely by water temperature.
- C) Higher temperature increases are generally associated with more severe bleaching and lower recovery rates, though the Red Sea appears more resilient than the pattern would predict.
- D) The Great Barrier Reef will lose all its coral within the next decade.

Question 10 (Rhetorical Synthesis - Easy)

While researching a topic, a student has taken the following notes: The Great Wall of China was built over approximately 2,000 years by multiple dynasties. The wall stretches over 13,000 miles across northern China. Its primary purpose was to protect Chinese states from nomadic invasions. Contrary to popular belief, the Great Wall is not visible from space with the naked eye. The student wants to correct a common misconception about the Great Wall. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The Great Wall of China is one of the most impressive architectural achievements in human history.
- B) Although the Great Wall of China stretches over 13,000 miles, it is not, contrary to popular belief, visible from space with the naked eye.
- C) The Great Wall of China was built over 2,000 years by multiple dynasties to protect against nomadic invasions.
- D) At over 13,000 miles long, the Great Wall of China is the longest structure ever built.

Question 11 (Command of Evidence: Textual - Easy)

An ecologist argues that urban green spaces improve mental health. Which evidence is most relevant?

- A) Green spaces improve property values.
- B) Cities have limited nature.
- C) Parks are nice to visit.
- D) A study using GPS data from 20,000 people found that spending just 120 minutes per week in green spaces was associated with 20% higher self-reported well-being, regardless of income, age, or exercise level.

Question 12 (Inferences - Hard)

Recent research on coral reef ecosystems reveals that bleaching events during which corals expel their symbiotic zooxanthellae algae under thermal stress have increased in frequency from roughly once every 25 years in the early 1980s to nearly once every 6 years by 2020. While some coral species can associate with more heat-tolerant algal strains after repeated thermal events, this "shuffling" mechanism requires multiple recovery cycles spanning several years and appears insufficient to keep pace with the accelerating rate of ocean warming. Which choice is most strongly supported by the text?

- A) Coral reefs in cooler waters are more vulnerable to bleaching than tropical reefs because they lack prior exposure to thermal stress.
- B) The chemical composition of warming ocean water, rather than the temperature increase itself, is the primary driver of coral bleaching.
- C) Coral species that experience moderate, non-lethal bleaching events at regular intervals may develop greater thermal resilience than those in consistently stable thermal environments.
- D) All coral species will eventually develop sufficient heat tolerance through algal shuffling to survive projected warming scenarios.

Question 13 (Inferences - Hard)

Archaeologists studying the transition from hunter-gatherer to agricultural societies have noted that early farming communities in the Fertile Crescent show evidence of significantly poorer dental health, shorter average stature, and higher rates of nutritional deficiencies compared to their foraging predecessors. Despite these apparent health costs, agricultural populations expanded rapidly, displacing hunter-gatherer groups across most of Eurasia within a few thousand years. Researchers attribute this expansion not to individual health advantages but to the dramatically higher population densities that sedentary agriculture could support. Which choice is most strongly supported by the text?

- A) The health costs of early agriculture were limited to dental problems and did not affect overall physical development.
- B) Hunter-gatherer societies voluntarily adopted agriculture because they recognized its long-term health benefits.
- C) Agricultural populations expanded primarily because their diet provided superior nutrition compared to foraging.

D) A subsistence strategy that is less beneficial for individual health can still confer a competitive advantage to a population if it supports substantially greater numbers of people.

Question 14 (Command of Evidence: Quantitative - Hard)

A scatter plot: $r=0.92$ between ice cream sales and drowning deaths. A politician proposes banning ice cream to save lives. The flaw:

- A) The correlation is too weak.
- B) The data is fabricated.
- C) Both variables are driven by a confounding variable (temperature/season), so the correlation is spurious and the policy would have no effect on drowning.
- D) Ice cream is unrelated to water.

Question 15 (Central Ideas and Details - Medium)

A passage describes how coral reefs, which cover less than 1% of the ocean floor, support 25% of all marine species. The author compares their loss to 'burning a library before reading the books,' emphasizing undiscovered medicinal compounds. The central idea is:

- A) The ocean floor is mostly empty.
- B) Coral reefs are beautiful.
- C) Libraries should not be burned.
- D) Coral reefs are disproportionately important to both biodiversity and potential human benefit, making their destruction a loss of irreplaceable scientific and medical knowledge.

Question 16 (Transitions - Easy)

During the Renaissance, wealthy patrons frequently commissioned portraits to display their status and power. To ensure they appeared appropriately majestic, patrons often demanded that painters ignore physical flaws and instead paint them with highly idealized features. some rebellious artists, like Caravaggio, pioneered 'chiaroscuro' and hyper-realism, choosing to paint their subjects warts, wrinkles, and all. Which choice completes the text with the most logical transition?

- A) Furthermore,
- B) Consequently,
- C) However,
- D) For example,

Question 17 (Transitions - Medium)

To ensure the structural integrity of the bridge during hurricane-force winds, the engineering team incorporated flexible joints that allow the roadbed to sway gently. rigid bridges are prone to snapping under extreme lateral pressure, this flexibility actually prevents catastrophic failure. Which choice completes the text with the most logical transition?

- A) Because
- B) Similarly,
- C) Therefore,
- D) Although

Question 18 (Boundaries - Hard)

The study identified four risk smoking, obesity, sedentary lifestyle, and excessive alcohol consumption. Which choice conforms to Standard English?

- A) factors;
- B) factors:
- C) factors
- D) factors,

Question 19 (Words in Context - Easy)

Despite the overwhelming data supporting climate change, a vocal minority continues to the scientific consensus. Which best fits?

- A) celebrate
- B) confirm
- C) dispute
- D) ignore

Question 20 (Form, Structure, and Sense - Easy)

Yesterday, the talented scientist a new element that amazed all the critics in attendance. Which choice completes the text so that it conforms to the conventions of Standard English?

- A) is creating
- B) created
- C) creates
- D) will create

Question 21 (Command of Evidence: Textual - Medium)

A nutritionist claims that intermittent fasting reduces chronic inflammation more effectively than a standard calorie-restricted diet. Which finding, if true, would most directly support this claim?

- A) A calorie-restricted diet is generally easier for patients to maintain over long periods than intermittent fasting.
- B) Both intermittent fasting and calorie restriction have been shown to promote weight loss.
- C) Intermittent fasting has become increasingly popular among celebrities and social media influencers.

D) Blood samples from subjects on an intermittent fasting regimen showed a 35% reduction in inflammatory biomarkers, compared to only a 10% reduction in the calorie-restricted group.

Question 22 (Command of Evidence: Quantitative - Easy)

A city planner surveyed residents about their preferred new public amenity. The options were a Park, a Library, or a Pool. The survey results showed a clear favorite among the residents. | Amenity | Votes |
|---|---| | Park | 480 | | Library | 220 | | Pool | 190 | Which choice best uses data from the table to complete the statement that the Park was the most popular option?

- A) The Park received 480 votes, which was higher than the 220 votes for the Library and the 190 votes for the Pool.
- B) The Library received 220 votes, which was the highest number of votes.
- C) The Pool received 190 votes, proving it was the most popular choice.
- D) The Park received 190 votes, making it the clear favorite among the options.

Module 2: Math

Question 23 (Equivalent expressions - Easy)

What is the GCF of 12 and 18?

- A) 36
- B) 12
- C) 3
- D) 6

Question 24 (Linear functions - Easy)

Parallel lines have slopes.

- A) opposite
- B) zero
- C) equal
- D) perpendicular

Question 25 (Linear equations in one variable - Easy)

Two numbers sum to 47. One is 13 more than the other. What is the larger number?

[Student Produced Response]

Question 26 (Two-variable data: models and scatterplots - Hard)

A researcher surveys 200 college students about their study habits and grades. She presents a scatter plot showing a positive correlation ($r = 0.72$) between hours spent studying and GPA. A skeptic argues that the correlation is misleading. Which of the following, if true, would most support the skeptic's concern?

- A) Students who study more also tend to attend class more regularly, and class attendance alone has a stronger correlation with GPA than study hours.
- B) The sample size of 200 is larger than most published studies on the topic.
- C) The scatter plot uses color-coded dots for each academic major.
- D) The researcher used GPA on a 4.0 scale rather than a 100-point scale.

Question 27 (Percentages - Medium)

A store sells an item for \$63 after a 30% discount. What was the original price?

[Student Produced Response]

Question 28 (Inference from sample statistics and margin of error - Hard)

A pharmaceutical company tests a new blood pressure medication. In a randomized controlled trial with 600 patients, the treatment group ($n=300$) shows a mean systolic blood pressure reduction of 11.2 mmHg, and the control group ($n=300$) shows a reduction of 7.8 mmHg. The 95% confidence interval for the difference (treatment control) is (1.1, 5.7) mmHg. The company must decide whether to proceed to Phase III trials, which requires the drug to show at least a 1 mmHg advantage over placebo. What does the confidence interval suggest?

- A) The drug should not proceed because the advantage could be as small as 1.1 mmHg.
- B) The drug should proceed because the entire confidence interval for the difference exceeds the 1 mmHg threshold, indicating the drug is both statistically and clinically effective.
- C) The drug should not proceed because individual patients may not all benefit.
- D) More research is needed because the confidence interval is too wide.

Question 29 (Two-variable data: models and scatterplots - Easy)

A scatterplot shows the relationship between the number of hours studied and exam scores. The line of best fit is given by $y = 6.5x + 40$. What does the 40 represent in this context?

- A) The expected exam score if a student studies for 0 hours.
- B) The expected increase in the exam score for each additional hour studied.
- C) The average exam score for all students in the class.
- D) The total number of hours a student needs to study to score 100.

Question 30 (Two-variable data: models and scatterplots - Easy)

A pie chart shows the allocation of a city's \$800,000 annual parks budget. Parks Maintenance is 45%, New Equipment is 25%, Staff Salaries is 20%, and Community Events is 10%. How many dollars are allocated to Staff Salaries?

[Student Produced Response]

Question 31 (Nonlinear equations in one variable and systems of equations in two variables - Medium)

Solve: $x^2 - 5x - 14 = 0$

- A) $x=7$ or $x=2$
- B) $x=-7$ or $x=2$
- C) $x=14$ or $x=-1$
- D) $x=7$ or $x=-2$

Question 32 (One-variable data: distributions and measures of center and spread - Easy)

Which can establish causation: experiment or observational study?

- A) Neither.
- B) Only observational.
- C) Only a well-designed experiment with random assignment.
- D) Both equally.

Question 33 (Linear equations in two variables - Medium)

Which of the following lines is parallel to $y = -\frac{3}{4}x + 5$?

- A) $y = -\frac{3}{4}x - 2$
- B) $y = \frac{4}{3}x + 5$
- C) $y = \frac{3}{4}x + 5$
- D) $y = -\frac{4}{3}x + 1$

Question 34 (Nonlinear equations in one variable and systems of equations in two variables - Medium)

What is the vertex of the parabola $y = 2x^2 - 12x + 22$?

- A) (3, 4)
- B) (3, 4)
- C) (6, 22)
- D) (3, 22)

Question 35 (Probability and conditional probability - Easy)

A number is drawn from 1-20. P(even AND greater than 10)?

- A) $\frac{3}{10}$
- B) $\frac{1}{5}$
- C) $\frac{1}{4}$
- D) $\frac{1}{2}$

Question 36 (Systems of two linear equations in two variables - Hard)

A system has no solution. The lines must be:

- A) Intersecting at the origin
- B) Identical
- C) Parallel with different y -intercepts
- D) Perpendicular

Question 37 (Area and volume - Easy)

A right circular cylinder has a volume of 45π cubic centimeters and a height of 5 centimeters. What is the radius of the base of the cylinder, in centimeters?

- A) 4.5
- B) 15
- C) 3
- D) 9

Question 38 (Nonlinear functions - Medium)

If $f(x) = 2^{x+1}$, which is equivalent?

- A) $2 \cdot 2^x$
- B) $2^x + 2$
- C) $2x + 1$
- D) $(2x)^1$

Question 39 (Linear functions - Medium)

Which of the following is an equation of the line that is perpendicular to $y = -\frac{2}{3}x + 4$ and passes through the point $(4, 1)$?

- A) $y = -\frac{2}{3}x + \frac{11}{3}$
- B) $y = \frac{3}{2}x - 5$
- C) $y = -\frac{3}{2}x + 7$

D) $y = \frac{3}{2}x + 1$

Question 40 (Nonlinear functions - Hard)

A radioactive substance decays according to $A(t) = A_0(0.5)^{\frac{t}{50}}$, where A_0 is the initial amount, t is time in years, and 50 is the half-life. If $A(t)$ is rewritten as $A(t) = A_0(c)^t$, what is the value of c ?

- A) $1 - \frac{0.5}{50}$
- B) $(0.5)^{\frac{1}{50}}$
- C) $(0.5)^{50}$
- D) $\frac{0.5}{50}$

Question 41 (Evaluating statistical claims: observational studies and experiments - Medium)

A hospital study found that patients who were prescribed a statin medication had better cardiovascular outcomes than patients who were not prescribed statins. However, a statistician pointed out that patients prescribed statins also had regular doctor visits, controlled diets, and exercise regimens. Which of the following best describes the statistician's concern?

- A) The statin medication may have side effects that were not measured
- B) The improved outcomes may be due to the regular doctor visits, diet, and exercise rather than the statins themselves, making these confounding variables
- C) The study should have included patients from multiple hospitals
- D) Statins are not effective for cardiovascular health

Question 42 (Evaluating statistical claims: observational studies and experiments - Medium)

A researcher found that among 5,000 employees surveyed at tech companies in Silicon Valley, those who reported working remotely at least 3 days per week had higher job satisfaction scores than those who worked entirely on-site. Which statement best represents what can be concluded from this study?

- A) Remote work causes higher job satisfaction for tech employees
- B) Among tech employees in Silicon Valley who were surveyed, remote work at least 3 days per week is associated with higher job satisfaction scores
- C) All employees should work remotely to increase their job satisfaction
- D) Remote work causes higher job satisfaction for all employees in all industries

Question 43 (Linear functions - Hard)

The value of a machine depreciates linearly from \$50,000 to \$14,000 over 9 years. What is the value after 6 years?

[Student Produced Response]

Question 44 (*Linear equations in one variable - Medium*)

A plumber charges \$75 per visit plus \$50/hour. If the total bill is \$275, how many hours did the plumber work?

[Student Produced Response]